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|-----------|---------------------------------------------------------------------------------------|-----------|
|           |                                                                                       |           |
| 4(a)(i)   | (stress =) force / cross-sectional area                                               | <b>B1</b> |
| 4(a)(ii)  | (strain =) extension / original length                                                | <b>B1</b> |
| 4(b)(i)   | $E = FL / Ax$                                                                         | <b>C1</b> |
|           | $= GL / A$                                                                            | <b>A1</b> |
| 4(b)(ii)  | straight line from origin above the original line                                     | <b>M1</b> |
|           | line ends at point (4 small squares, $F_1$ ).                                         | <b>A1</b> |
| 4(b)(iii) | 1. shaded area below the graph line and between the two vertical dashed lines         | <b>B1</b> |
|           | 2. remove the force/ $F/F_2$ and the wire goes back to original length/zero extension | <b>B1</b> |
| 4(b)(iv)  | values have a large range                                                             | <b>B1</b> |

| Question | Answer | Marks |
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